

```
C exp31.c
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1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main()
5 {
6     printf("=====\n");
7     printf("      ARP TABLE - EXPERIMENT 31\n");
8     printf("=====\n\n");
9
10    printf("Displaying ARP table...\n\n");
11
12    system("arp -a");
13
14    return 0;
15 }

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PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> .\exp31.exe
Displaying ARP table...

Interface: 10.48.30.204 --- 0xf
Internet Address      Physical Address      Type
10.48.30.80           6e-07-9c-f4-16-a8     dynamic
10.48.30.255          ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.24.112.1 --- 0x30
Internet Address      Physical Address      Type
172.24.127.58         00-15-5d-90-e8-44     dynamic
172.24.127.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
239.255.255.250       01-00-5e-7f-ff-fa     static
```

```
C server27.c
C server27.c
1 #include <stdio.h>
2 #include <string.h>
3 #include <unistd.h>
4
5 #pragma comment(lib, "ws2_32.lib")
6
7 int main()
8 {
9     WSADATA wsa;
10    SOCKET serversocket;
11    struct sockaddr_in server, client;
12    int clientlength = sizeof(client);
13
14    char domain[100];
15    char result[100];
16
17    // DNS records
18    char *domains[] = {
19        "google.com",
20        "example.com",
21        "localhost"
22    };
23
24    ...

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PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> gcc client27.c -o client27.exe
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> gcc server27.c -o server27.exe
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> .\client27.exe
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> .\server27.exe

Enter domain name: google.com
IP Address : 142.250.72.14
% PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB>

DNS Server started...
Waiting for DNS query...
Query received : google.com
Response sent : 142.250.72.14
% PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB>
```

```
C exp32.c
C exp32.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char data[100];
7     char stuffed[200];
8
9     int i;
10    int j = 0;
11    int count = 0;
12
13    printf("=====\n");
14    printf("      BIT STUFFING\n");
15    printf("=====\n\n");
16
17    printf("Enter bit string: ");
18    scanf("%99s", data);
19
20    for (i = 0; data[i] != '\0'; i++)
21    {
22        stuffed[j++] = data[i];
23
24        if (data[i] == '1')
25            count++;
26    }
27
28    stuffed[j] = '\0';
29
30    printf("Original Data : %s\n", data);
31    printf("Stuffed Data : %s\n", stuffed);
32
33    return 0;
34 }

Problems Output Debug Console Terminal Ports

PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> gcc exp32.c -o exp32.exe
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> .\exp32.exe

=====
BIT STUFFING
=====

Enter bit string: 01111110111111

Original Data : 01111110111111
Stuffed Data : 01111110011111101011111010111110
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB>
```

```
C exp28.c
C exp28.c
1 #include <stdio.h>
2 #include <winsock2.h>
3 #include <ws2tcpip.h>
4
5 #pragma comment(lib, "ws2_32.lib")
6
7 int main()
8 {
9     WSADATA wsa;
10    char hostname[100];
11    struct hostent *host;
12    char ip[INET_ADDRSTRLEN];
13
14    // Initialize Winsock
15    if (WSAStartup(MAKEWORD(2, 2), &wsa) != 0)
16    {
17        printf("Winsock initialization failed.\n");
18        return 1;
19    }
20
21    printf("Enter hostname: ");
22    scanf("%99s", hostname);
23
24    // Resolve hostname
25    host = gethostbyname(hostname);
26
27    if (host == NULL)
28    {
29        printf("Invalid hostname.\n");
30        return 1;
31    }
32
33    printf("Host Name : %s\n", host->h_name);
34    printf("Address Type : %s\n", host->h_addrtype);
35    printf("Address Length : %d\n", host->h_addr_len);
36    printf("IP Address : %s\n", inet_ntoa(*(struct in_addr *)host->h_addr));
37
38    return 0;
39 }

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PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> gcc exp28.c -o exp28.exe -lws2_32
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB> .\exp28.exe
Enter hostname: google.com

Host Name : google.com
Address Type : IPv4
Address Length : 4
IP Address : 172.217.24.174
PS C:\Users\DIVAKAR V\Downloads\cn\CN LAB>
```